Exercise - 5.3

1. Find the sum of the first 20-terms of the arithmetic progression having the sum of first 10 terms as 52 and the sum of the first 15 terms as 77.
2. Find the sum up to the 17th term of the series $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \cdots$
3. Compute the sum of first n terms of the following series:

   i. $8 + 88 + 888 + 8888 + \cdots$ ii. $6 + 66 + 666 + 6666 + \cdots$

4. Compute the sum of first n terms of $1 + (1+4) + (1+4+4^2) + (1+4+4^2+4^3) + \cdots$
5. Find the general term and sum to n terms of the sequence $1, \frac{4}{3}, \frac{7}{9}, \frac{10}{27}, \ldots$.
6. Find the value of n, if the sum to n terms of the series $\sqrt{3} + \sqrt{75} + \sqrt{243} + \cdots$ is $435\sqrt{3}$.
7. Show that the sum of $(m+n)^{th}$ and $(m-n)^{th}$ term of an AP. is equal to twice the mth term.
8. A man repays an amount of Rs.3250 by paying Rs.20 in the first month and then increases the payment by Rs.15 per month. How long will it take him to clear the amount?
9. In a race, 20 balls are placed in a line at intervals of 4 meters, with the first ball 24 meters away from the starting point. A contestant is required to bring the balls back to the starting place one at a time. How far would the contestant run to bring back all balls?
10. The number of bacteria in a certain culture doubles every hour. If there were 30 bacteria present in the culture originally, how many bacteria will be present at the end of 2nd hour, 4th hour and n^{th} hour?
11. What will Rs.500 amounts to in 10 years after its deposit in a bank which pays annual interest rate of 10% compounded annually?
12. In a certain town, a viral disease caused severe health hazards upon its people disturbing their normal life. It was found that on each day, the virus which caused the disease spread in Geometric Progression. The amount of infectious virus particle gets doubled each day, being 5 particles on the first day. Find the day when the infectious virus particles just grow over 1,50,000 units?

5.5.3 Some Special Finite Series

In this section we give some of the important formulas of summing up finitely many terms which follows either an AP, GP, or any specific series.

1. Summation of first n natural numbers:

$$\sum_{k=1}^{n} k = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

   [Treating this as an AP, one can find the sum.]
2. Summation of the squares of first n natural numbers:

$$\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

   [Use the identity $a^3 - b^3 = (a-b)(a^2+ab+b^2)$ and try to prove this result.]
3. Summation of the cubes of first n natural numbers:

$$\sum_{k=1}^{n} k^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

   [Use the identity $k^4 - (k-1)^4 = (4k^3 - 6k^2 + 4k - 1)$ and try to prove this result.]

Note that the above three results were proved in the earlier classes.